

Project Planning Phase

Project Planning Template (Product Backlog, Sprint Planning, Stories, Story points)

Date	15 February 2026
Team ID	LTVIP2026TMIDS81120
Project Name	Smart Sorting: Transfer Learning for Identifying Rotten Fruits and Vegetables
Maximum Marks	5 Marks

Project Planning

The development of the Fruit Freshness Classification System was carried out in a structured and systematic manner to ensure accuracy, reliability, and practical usability. The project planning phase included problem analysis, dataset preparation, model development, evaluation, and deployment.

1. Requirement Analysis

The first stage involved understanding the challenge of identifying fruit freshness using visual inspection. Based on research, key visual attributes such as color variation, texture changes, and surface patterns were selected as important indicators. The objective was defined as building an image classification model capable of predicting whether a fruit is Fresh or Spoiled.

2. Data Collection

A labeled dataset of fruit images was collected from reliable open-source platforms. The dataset contained images categorized into freshness classes. Images were organized into structured training and testing folders to support efficient model training and validation.

3. Data Cleaning

Data cleaning ensured that all images were valid, properly labeled, and readable. Corrupted or unsupported image files were identified and removed. Folder organization was verified to prevent class mismatches. This step guaranteed dataset consistency before model training.

4. Data Preprocessing

Preprocessing prepared images for deep learning. All images were resized to a uniform dimension, normalized for consistent pixel values, and augmented using techniques such as rotation, zooming, and flipping to improve model generalization. The dataset was divided into training and validation sets to evaluate model performance.

5. Exploratory Data Analysis (EDA)

Exploratory analysis involved visualizing sample images and examining class distribution. Training metrics such as accuracy and loss were monitored to understand learning behavior. This stage helped confirm dataset balance and identify potential overfitting issues.

6. Model Development

A convolutional neural network was designed and trained to learn distinguishing visual features between fresh and spoiled fruits. Different training configurations and augmentation strategies were tested to improve model performance.

7. Model Evaluation and Selection

Model performance was evaluated using validation accuracy and loss trends. The best-performing configuration was selected based on prediction consistency and generalization capability.

8. Model Deployment

The trained model was saved and integrated into a Flask-based web application. The system allows users to upload fruit images and instantly receive freshness predictions through a simple interface.

9. Testing and Validation

The application was tested with various fruit images to ensure stable predictions and smooth interaction. Edge cases such as unusual lighting or image orientation were verified to confirm reliability.

This structured planning ensured the fruit freshness classification system was developed methodically, with strong emphasis on dataset quality, model accuracy, and real-world usability.